

Active Inference Stream #009.2: "The Projective Consciousness Model and

2:04:16

hello everyone welcome to the active inference live stream it is ACM stream 9.2 and it is December 1st 2020 so happy December to everyone today we are really excited to have David um one of the authors of the paper we're going to be discussing so it will be just a great discussion today really looking forward to it welcome to teamc everyone tecom is an experiment in online team communication learning and practice related to active inference you can find us on our website at Twitter via email at our YouTube channel or our public keybase team and keybase username this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work also all perspectives and backgrounds are welcome here and as far as video etiquette for live stream go mute if there's noise in your background raise your hands so that we can hear from everybody in the conversation and let's use respectful speech Behavior just some quick updates on upcoming ACM streams today we are in 9.2 December 1st talking about this projective Consciousness model paper and in the rest of December we'll be talking about two papers on December 8th and 15th will be in ACM 10 on a variational approach to scripts and on December December 22nd and 29th will be in ACM 11 on sophisticated effective inference so

two cool and very interesting papers that we'll be getting to but today in act M9 we're still continuing our discussion of projective Consciousness model first we're going to have some introductions and warm-ups then we'll go through 9.2 as with previous times the projective Consciousness model paper by Will aord at all 2018 and in last conversation in 9.1 1 and 9.0 we talked about some of the groundwork and the aims and claims of the paper as well as going through the abstract and the road map today especially because we're lucky enough to have an author on board we're going to be talking about a few deeper and auxiliary questions as well as going through a few key quotations and talking about maybe some of the implications for the paper and then as always we'll return to the figures and ask what each figure is representing or invoking so cool to have this discussion today and then different papers in 10 and 11 and then we'll have a whole new suite of papers in 2021 so get in touch with us if you want to participate first though with our nice small group today let's introduce ourselves we'll go around and give a short introduction or a check-in and then just pass it to somebody who has not spoken yet so I'm Daniel I'm in California and I will pass it to Adam uh hi um I'm Adam uh Adam saffron I'm in uh Bloomington Indiana at Indiana University as a post-doctoral researcher and um I'm very interested in Consciousness and uh recently came up

with a theory I called integrated World modeling Theory which has uh a lot of overlap with uh David's work and which I find to be really fascinating and inspiring so glad to be here cool let's go to David hello there I'm David rud I'm associate professor at the University of Geneva Switzerland uh and I'm one of the coor of this paper and somebody work on trying to model the role of Consciousness including it phenomenal aspects uh in let's put it that way generally speaking biological cybernetics and more particularly active inflence thank you for having me here today cool let's go to Sasha welcome Sasha hi everybody um my name is Sasha I'm in Davis California and um I study developmental neuroscience and how um synapses form in the developing brain and um really interested to be part of this conversation today cool then we'll go Alex and Stephen hi everybody I'm Alex I'm in Moscow Russia and I'm a researcher in systems management school and I pass it to Steven hello I'm Stephen I'm based in Toronto and I'm doing a practice based PhD in to uh particip atory theater approaches and the use of Landscapes to explore themes and back to Daniel I believe that introduces our nice group today as far as the warm-up questions go we'll just have people raise their hand whenever they feel like it and I'll just drop out

the first two questions at the beginning
and they are what interest you about
studying Consciousness and what is a way
that we can study it so both David and
Adam I know you've thought a lot about
this maybe one of you two would want to
go first um

here yeah David just how how did you get
to this uh area it's a it's just
something it be a long story but to make
it short uh let's put it that way it's
really related to philosophical and
existential question and this the Quest
for turning this type of of question
into something scientific um but
actually I was not that interested in
Psychology when I was younger I wanted
to be either an astrophysicist or a
conductor you know of Orchestra but I
did

not interesting it's a different type of
coordination that you're achieving now
uh Stephen and then anyone else who
wants to raise their

hand yeah I'm I'm interested in
conscious

it's in a way it's kind of a late
arrival because I um I always thought
that was something esoteric that I
couldn't really participate in but um
and uh I've suddenly seen this has start
to come Center Stage um and in both this
active inference world but also being
talked about more in in in in the
participatory theater environment so you
know what what does it mean to be
consciously aware to have imagination
and what I

think is quite interesting is to try and
move Beyond this kind of pan
psychic um approach that seems to
permeate um a lot of people who actually
go there um either it's either something
that's ignored and pushed to the side or
it's everywhere and everything in this
pan psychic and what I am really
interested is like how can it be engaged
in a kind of in terms of how we are
reflective or reflexive or participatory
and what Consciousness means in terms of
talking to people who aren't the same as
us interesting definitely the coming
from a different perspective is
something that's baked into this type of
a framing because yeah everyone's coming
from a different perspective that's our
starting point that's not our conclusion
here and then there's questions about
how we're going to coordinate amidst
that so we can either have anyone else
raise their hand on these questions or
I'll just put up this special point²
question which is what is something that
you have thought about differently after
reading the paper or participating in
last week's discussion and for me it was
actually uh biking in the morning and
seeing the moon just look massive and
thinking about that illusion in the
context of the projection and what we
talked about last week as well as some
other literature by these authors so I
just thought what might be some
differentiable experiences some little
tests that we could do that would be
able to capture differences between the

projective model and other models of
Consciousness I didn't really have an
answer but it was just something I was
thinking about after our discussion
anyone else um after reading the paper
or participating in the discussion
Stephen well one thing that it was
really helpful this unpacking the
diagrams last week and I noticed that
there was that box for the head on of of
of the perspective the Consciousness
perspectival Consciousness and it
suddenly made me realize that yes this
can Bridge more back into the embodied
work because it I had a discussion about
this paper before with someone who's in
theater and interested in kind of some
of this more kind of esoterical kind of
ways of knowing but I'm trying to find a
way to bridge it back to the embodied
sort of inactive work and I sort of by
seeing it in the in the kind of head I
suddenly realize okay it still does come
back to the physicality so that was
really really useful for me
cool cool

Sasha um yeah after last week I just
began seeing everything as um you know
more more geometrical and thinking about
um uh people and their behavior as
projections um kind of yeah um
recognizing that we're always seeing
just one side or one perspective of
someone's behavior and um specifically
that that GB logo it just uh that's all
I could
see right with light from the different
side showing the different letters

GB yeah and we'll come back to GB a little bit later in this discussion any other thoughts from the panelists on these warm-up questions otherwise I'm sure there'll be a lot to get into a lot of tangents to go off on to cool all right David go ahead no I would just I just wanted to say that there would be a lot to say about the points and questions that were raised but perhaps we can not do that now and go through the the the stream and and see your questions come about great so today in this discussion as with our last two weeks we've been talking about the projective Consciousness model and phenomenal selfhood which is a paper in Frontiers and psychology December 2018 by Will aord at all and thanks David again for coming out to the stream it's always awesome to have the authors the aims and claims of the paper we did go through in a previous week but David just I'm curious here uh how do you present this work or how do you introduce this topic to different audiences who might have familiarity with different topics like how do you frame the aims and the claims of the paper so lately I've been particularly talking in context where you know people come from artificial intelligence robotics virtual reality um with you know of course some more broader philosophical interest and I generally try to frame it starting bit like you did I mean from

the question of you know what we could call biological cybernetics and you elaborated on that term before uh and how active inference for situated autonomous system you know with survival challenges is an interesting paring to understand cognition within biological cybernetics and then I come to the question of what would be then the the role in that process of what you know we in it as being Consciousness what we can say about it emphasizing the the big and important challenge of connecting let's put it that way for short you know the phenomenology of subjective experience is some aspect of it with for lack of a better word functions uh why do we perceive and imagine the world the way and others and ourself in the world the way we seem to do at a certain level and uh how that actually how can we make sense of that to understand computational in some sense process that are key to you know adapt adaptation uh you know resilience in the context of biological cybernetics cool thanks for that and I think it was really clear how you laid out that connection in the paper basically there's the functions of Consciousness which we'll go through soon and then there's the uh subjective aspects which were captured in the invariance and the PCM is like a bridge that's going to be connecting the subjective invariance of Consciousness to some of these functional attributes and so we're kind of going from point A

to point B uh across a specific River
and so if we can really be clear about
which sites on the riverbank we're
building from and then be really clear
about what the bridge is and isn't then
we'll have a really good discussion
about Consciousness as opposed to just
swirling around and thinking it does
everything and nothing and it's
everything and nothing at once and all
these things so any thoughts on the aims
and claims from anyone else otherwise
we'll just go through to the next
section cool yep we have a lot of
questions coming up so in uh previous
weeks we looked through the abstract
sentence by sentence just to understand
how the authors were taking the free
energy energy minimizing approach of
active inference biological cybernetics
and then connecting this to a few other
areas such as geometry and the embedded
psychology and uh that was the abstract
in the road map we had the sort of
outline of where they go they start with
an introduction and go straight to the
methodology and then talk about the two
river banks in that previous metaphor
that we want to connect the
phenomenological invariance of
experience and then the functional
features of Consciousness and then sets
up the problem in a way that hopefully
could be solved so maybe uh David I'm
curious like in other papers how have
you structured it differently or this
was in 2018 what have you been um like
wishing that could have been explained

more or where have have you bolstered the theory since then all right that's a huge uh question um so number one kenet Williford the first author of this paper we expected him to join I don't know if he will join later is a friend of mine and I admired for the quality of his mind is a philosopher and so the structure of this particular paper we look at today is I think deeply informed by his you know philosophical methods uh if you were to read the first paper we published on the topics in the Journal of biological theoretical biology sorry uh in 2017 you you would find a two wordy though but very different way of presenting but they are totally consistent with each other and uh in both case you know it's still work in progress so there are still a lot of question question and of course we are not claiming that we explain everything by far we're trying to restrict ourself to things we think we can frame in a classical scientific framework and you know I've been working for years now in my lab and we are actually finishing a paper on that to integrate in this in the general based on the general principle of this model in addition to um visual illusion as we did recently in the Moon Illusion paper that was published in 2020 uh mechanism of effective and social appraisal and processing and social perspective taking based on projective geometry notably but not only um to model and understand how

artificial agent virtual or robotics as
a model of ourself if you wish can do
theory of mind and how that plays role
in Emotion regulation uh adaptation of
behavior um and that's basically what
I've been doing and in interestingly
without taking too long there are some
studies from psychophysics that that
show or that you know based on empirical
data
that if you are confronted with an
object or something you you you feel
is dangerous like inance a snake or for
some phobic for instance uh the
intensity of your fear will be a a
function of the
inverse more or less the inverse of the
distance of the object and that has been
formalized as being you know something
you can identify with an empirical
psychophysical law called Stevens Law
with a certain type of exponent and as
it turns out actually the projective
geometrical framing just give you that
naturally it transform information
notably the size the apparent size of
object according under certain
constraint to AO in that is inverse from
the distance more or less so you can
recover things that have been documented
before and understand how they may play
a role in appraisal something I feel
when it's closer it's bigger it takes
more of my field of Consciousness if you
wish to make it simple and so one way of
motivating my behavior of for instance
avoidance is imagining myself farther so
the thing is smaller takes less space

and as a result it has less impact following the very same law we were discussing and therefore that will minimize my free energy and I will move away and I do that with robots and things like that huh very cool yeah I'm imagining this like fear is proportional to one over distance something in that family of equations uh like a bus that's uh 100 feet away not scary 50 feet away not scary it's still on the flat part of the fear curve and then it gets exponentially higher when it gets closer to you and takes up more of your field of vision so um and then there's also longdistance threats somebody with a projectile or something like that and so that entails a causal model where somebody who's doing a certain action might be able to make an object quite big in your face very soon but maybe it's not quite there yet but it could happen rapidly so that's kind of an interesting take yeah so the point here is in along the line of what they were saying trying to understand what is or what could be some aspect of the causal role of Consciousness in things like appraisal and motivation of behavior here the point that we make is that actually the spatial structuring that we put forth based on projective geometry implies naturally something about you know relative size that is consistent with psychophysics and so it's to say that find somehow this Con structuration of space by conscious

act as a sort of lens a magnifying device if you think about it that way that in virtue of its own geometry will have an impact that you know can then be used for other aspect of motivation and programming of behavior in the context of active

influence so maybe we'll get back to this Intermodal area again but that's the visual field and uh not all organisms or humans may have the visual field but it's clear that the spatial objects we visualize them in space and uh there's something similar though like if there's a lot of noise in the background but something is what you're taking up your field of audio awareness and so it's like in each of our senses there can be a sense like a sensory um expansion of focus greetings Lee just play on we're just continuing the discussion but go ahead there

David yes so one of the essential thing we we emphasize is that that's it's actually not just the visual field it's the C it is the cognition of space as in form in a multimodal manner you know through audio you may recognize object and all of a sudden you kind of have an image of sort the spatial representation of that object as if you could see it even though you don't see it blind people do that all the time they have actually perspectival representation and represent object in space based on other sensory channels there is an infer that is being made and the idea is that Consciousness operate

at some level of abstractions once you have actually extracted information uh from different sense modalities and then they become spatial object if you wish with variety of properties that you put in perspective in that field of Consciousness and based on which you will operate at the conscious level if that makes any sense yep interesting it's like it's so hard sometimes to disentangle from the eye eyes Vision perception debacle nightmare because we know that we're not experiencing what photons hit our retina we know it's a generative model and you're really taking it to the next level and it's like no you're not even just seeing a generative model of the visual field you're seeing this extremely integrated field of Consciousness and that um may depend on your visual abilities or your Acuity but it also something that's quite uh derived from Intermodal stimuli and uh yeah well that's cool stuff let's get to this projective geometry idea and we as non- geometers talked about it a little bit in previous discussions and just try to capture a few of these points that you said were differing critically between ukian and non- ukian geometries so I'm just curious and perhaps those who are not familiar with hearing about geometry and Consciousness in the same sent just what is it that brings geometry to the table and how do we end up even talking about geometry and Consciousness

together excellent question so first of all a little disclaimer I'm not a mathematician even though since I understood early on that what I was trying to do required to think mathematically in Fairly Advanced way banged my head to try to progress but I'm privileged to work with real and very you know good uh mathematicians so that helps but nevertheless I can give you to the limit of my knowledge a some info about it number one I would say generally speaking uh you know if Experience One conspicuous aspect of experience of subjective experience is a spatial manifestation not all experience reduced to space but a big chunk of it is involved it so it's good from a scientific standpoint because then it becomes something you can think about mathematically and a lot of models in let's put it that way in cognitive science has been based on information processing sometime very formal model like you know Advanced basion models or even active inference and so there is a lot of emphasis on the mathematics of information Theory so to speak and here the notion is that in order to bridge um information processing if you wish with aspect of experience as manifested as a space geometry comes obviously as a scientific object as a mathematical object as something you know necessary so the the the challenge is then to understand how to articulate information Theory with geometry in order to have a

fuller account of you know the process
so then now on the U specifics of
projective geometry it start from you
know trying to find some let's call it
that way phenomenological invariant that
most people would agree upon to a
certain extent considering experience
introspectively it appears that it's
obvious in Vision but I claim it's
General in both perception and
Imagination that the way we quote
unquote represent the world let's not
you know enter into the metaphysics of
representation let's use it in a
pragmatical way here this notion of
representation uh is in perspective you
know obviously a parallel line do not
appear to us in Vision as parallel but
they converge at a certain vanishing
point toward Infinity um there is an
horizon and it's three-dimensional it's
not like just the projection of a 3D
World on an image we experience it with
a sense of depth and I argue that the
same when you imagine yourself
somewhere considering something aiming
at something you know the notion of
aiming at is very important in the
phenomenological tradition in philosophy
it's not just about visual Vision it's
about spatial aiming and in all the
circumstances the world is mapped is
there a world that seems to correspond
to surrounding a world that corresponds
to thing we remember or even a totally
fantasy world of things that may never
exist every single time we
build experience in Consciousness the

point is that there is a perspective that is being taken with these properties of having an elusive point of view somewhat in perception we tend to feel it in we feel that we are in the middle of that space so that's the origin is somewhere in your head perhaps but that's not necessarily the case uh and that's not something that we can uh identify as an the way we could identify the location of an object in space it's illusive something that the phenomenologists have talked about a lot and it's in perspective so if you take those two things An Origin of the projection or of the point of view that is Elusive and a three-dimensional representation of space that is in perspective then the only geometry and the the geometry that account for that is projective geometry and here since we say the space is in three dimension uh there is this notion of you know left right up down and and forward and backward uh it's a three-dimensional projective space and then that means that this space is ruled by the So-Cal projective transformation that are specific type of transformation that comes from a group of transformation that transform perspective into that space uh so for instance you can transform a point of view uh uh from the point of view I have here looking at that screen to an imaginary point of view I would have in a different position myself or somebody else and all of that is naturally

governed by that geometry so then that geometry appears naturally as a or if not D candidate to contr this aspect of phenomenological experience and therefore then you can start thinking about okay how that can play a role in the process of active influence nice thank you for the clear explanation um really in in that uh little bit the clearest part moving forward is this elusive origin and spatial perspective that we have if someone can imagine an experience that doesn't fulfill those two conditions then we're already breaking our assumptions but if you can that's great it's like write a paper about it but given that there's consensus that there's an elusive origin and a spatial perspective we can then say that we're in a family of uh geometries that are related to this projective transformation and so going from there there's other Downstream consequences so really difficult topic sometimes to talk about of course transdisciplinary ideas but I think it's really clear how the structure of your argument is set up Stephen and just a couple of questions is is the is the um the distance as things go away is that and you feel less intensity in terms of the fear or something is that linear or is that kind of exponential or is it veryy depending on the space and and I suppose the other question is is the is is this transformation between one mapping to the kind of projective geometry it's so

it's it's not like you store it in that form there's some sort of action to make it into this projection yes very good question I'm going to try to address them to the extent I can so number number one these experiments that shows the spatial relationship between subjective fear and distance they found is a power law with an exponent minus one so basically it's like the fear decreases as a function of the inverse one over the distance so it's not linear it's not exactly an exponential it's not linear and uh as it turns out that's the very basic rule of perspective when you want to build a space in perspective you divide by your coordinates XYZ coordinates uh by a parameter that corresponds to the distance along with you are basically projecting the direction of aiming that can be for instance in front of you so basically you find that in that geometry there is a similar operations if you wish as those that are implied by the the one of distance models that is empirically derived so that's nonlinear but it's a very particular type of nonlinearity that is not a super complex nonlinearity that can be handled computationally which I guess is is important and now for your the second part of your question let me can you can you remind me of it so that I well it's just about that transformation between however the brain stores it and this

projective phenomenon that we experience yes so the idea here somewhat what we have to presuppose if we want to embrace that type of model is that there is an unconscious World model which is complex which is constantly updated that somehow store information more or less in memory uh in Clan manner uh uh and uh uh and that we access that information by taking perspective on it through by transforming this sort of quote unquote a clan representation into a perspectival objective representation that involve multiple steps and that's something for instance we develop the most basic one I would say uh in our mon vision paper in which we show uh and we how the calibration of a projective frame me 3D so we set the parameter that defines the projection under free energy minimization that's the idea is free energy minimization acting on Geometry uh UNS Illusions such as the Moon Illusion explain illusion such as aim illusion and many other illusion we would not develop in this paper and that is somewhat the idea is that in you know sometime people told talk about access Consciousness you know let's not enter into the complexity of that debate but let's say that the hypothesis here is that in the way we access information that's in Consciousness there is somewhat a change of coordinates quot quote from a edian representation to a projective one and that operation is essential in what

makes information conscious versus not
you know model if makes any sense thanks
for the answer let's go to
Lee oh hi yeah so I was just wondering
um to to sort of build on Steph's
question um if if a fear response is uh
exponentially related to distance does
that hold or is there any research that
shows whether that holds for a
counterfactual version so I'm thinking
here about phobia or trauma so if you
have a particular experience you know
often in PTSD those things are replayed
um so yeah so did you have any thoughts
on that or any pointers it's really a
great question
uh okay the literature on on this on the
the basic experiment is not huge
surprisingly and this and your and your
question is super interesting indeed
because we would like to to demonstrate
that even if you do it in an imaginary
way you cover this type of
effect consistent with the model I'm not
aware of such study but I'm you know I'm
doing so so much multidisciplinary work
that the literature I have to cover is
vast and sometime I may miss important
studies in specific subfield but
definitely that's you know that gives us
the road map for a great experiment
yeah in in the in some of the literature
I'm thinking of um
Benjamin Bergen on simulation he talks
about the difference between um
canonical perspective and I can't
remember the other one but it's
essentially firsters perspective um and

and the ability to shift between these
so I just wondered whether it might be
related but I'd be certainly interested
if there's if there's any research that
you're thinking about
doing yeah um David unmute and then go
ahead and then Sasha
so you want me to speak now oh if you
have anything to add there no no noit
sorry I misunderstood I lost the screen
oh yeah no great Point um and thanks for
that question Lee um
Sasha um yeah very interesting stuff um
makes me think about
um uh this Consciousness and
understanding of
um how to interact with your space as
like narrative building and and how that
develops over the lifetime that um like
as a child that the things that you're
afraid of are uh quite specific and
nearby um and as you know as we grow up
um we're afraid of much kind of larger
but more distant things but uh there's
something you know uh some
transformation we have to figure out as
far as um we still have to be motivated
and um afraid of the things that are uh
either very afraid or very or sorry very
scary or very unlikely and those are
still enough to motivate um people to
avoid them and so I think that's a
really interesting uh yeah something in
there about uh changing the scope of
your uh non- ukian projection let's say
um as you uh grow up nice Sasha and it's
kind of like when you go back to see
something scary in childhood you go oh

wait it was only a 6in stair that we
were jumping off of well you're bigger
but also you have a different worldview
and then just like in the computational
Psychiatry realm we might talk about
difference in a statistical parameter
leading to decreased motivation or
decision- making it could be a
distortion of this projection that
results in things being uh perceived as
dangerously close claustrophobia people
are too close to me or people are too
far away from me there's so much there
with a social touch as well um let's go
just on to the sort of new parts of 9.2
and talk about some of these other
slides and questions we have so last
week we talked about the functional
features the cybernetic features as well
as some of these other theories threads
that were drawn in like the availability
kind of like a workspace the simulation
Theory motivation of action uh we talked
about that last week as well as the
questions about the phenomenological
invariance with the emphasis on 01 being
that relational phenomenal
intentionality so just pause the video
and look at those if you want more
detail or of course read the paper but
let's go to some feedback from
participants so one thing that a
participant who couldn't be here today
asked was one thing I'm curious about is
whether they're working on more complex
forms of the model with higher temporal
depth that kind of speaks to these adult
fears considerations we talked about and

then also this person added another thing I'm curious about more philosophically is what do they mean by now they have an extended phenomenological background for the model which uh suggests that the now would be extended like an extended now however in naturalist models which I assumed uh active inference or potentially artificial intelligence framework based models are a subset of time is often modeled and understood as point- like or as a continuous variable with points I wonder what the implications are or what there might be if they mixed up these two interpretations of what the now is the physical and the psychological now so David what do you think about these questions from the participants oh they are great uh the I'm going to try to address some of it of course and again it's there are still a lot of things to discover and understand and one of the things we like about starting with a mathematical model is that then it acts as a guide now it's the theory that tells us you know where are the problems and where are the solutions if you have a general analogy sorry I'm going to be a bit off track but it's I think useful and without comparing ourselves with Einstein don't worry we don't have that you know ambition uh if you think about uh general relativity once the theory was derived the mathematical theory was derived basically people looked into it and

started thinking H there is a problem with this Theory it tells us that there should be black holes right and in fact basically the theory predicted things that you know nobody could observe of course before and very difficult to observe and that was a consequence of the mathematics and so that's the idea that we follow from a scientific standpoint is let's give us mathematics and see where it lead us now um one thing I'm curious about the the first question yes we are working on more complex form alluded to that at the beginning uh number one this notion of transformation of as an inverse law of distance related to this literature is not to say that we praise information only that way the it was to show at least to that type of relation hold in appraisal and to some extent or model makes sense of it now about the more complex form remember the space here the geometry is one thing but it's embedded into an active inference framework with a lot of complex computation involving prior beliefs and preference internal model of the body of others of objects including internal model that's what we are doing now of you know beliefs about the beliefs of others and beliefs about the beliefs that other belief are sorry beliefs about the belief that other believe other have so basically level six or second order theory of mind and by actually imbuing artificial agent with

this type

of statistical

processing uh and combined with the geometry then we can start modeling how you know inspired from simulation Theory we can put ourself in the shoes of other we can try to imagine what it would feel like to look at things to observe things to might aim at things as they seems to do

including understanding that they may have different beliefs and preference as us you know the same object may look cool to us and so when it get closer it's better but for them maybe that's something they hate and when it get closer they know they want to run away so then we have a way of extending the model very naturally with the same match with the same general principle to encompass theory of mind and social affective perspective taking if you wish and one interesting things that we find is that when you enable artificial agent with this type of process then they and you enable them also to be influenced by each other because you may be able to understand and predict other Behavior but not care about it but if you do then the agents the little robots or the virtual humans start Dem showing joint attention Behavior they start interacting looking at the same thing looking at each other to kind of confirm that they understand what the other likes and they constantly put themselves in the shoes of the other and reciprocally and through this process of

you know projective perspective taking
uh informed by all these kind of ban
like models under free energy
minimization they start showing very
complex Behavior including you know we
have reproduced in a nonverbal way the
famous San An task we know about the of
mine basically that's two puppet one of
them they are together there is some
object of interest in a basket one of
them leave the other take the object of
interest from the basket and put it in
the box and typically you ask children
you know when the other come back you
know where do you think you know she
will go and if you understand false
belief you will say well because she
believed that the box is still in the
the the object is still in the basket
she will go to the basket and so we have
model the kind of simulation framework
in a nonverbal way for that and we show
agent actually exploit their
understanding of false belief in others
to approach
disingenuously uh object they would like
by you know being freed of being buzzed
by the other because now the other has
false belief the same way we can kind of
simulate some aspect of social anxiety
because anytime you give you have an
agent that believe that the other agent
doesn't like it you know uh then you
know playing the algorithm just running
the algorism anytime the agent imagine
itself approaching the agent the other
agent that the agent suppose do not like
him then it put itself because he

practice he does social perspective
taking in the shoes of this o agent and
when you imagine what it is like for
this o agent to see himself or itself
get close
because of the same rule we have
discussed he understand that would raise
free energy in the other agent and since
he takes into account the oent opinion
that it attributes to the others
therefore that raise its own free energy
as a result avoid it tend to avoid
approaching the agent that it believe
doesn't like it so you see you start
having very sophisticated behavior all
through this basic uh processing so
that's for the first question sorry it
was long I hope it was not too confusing
anytime you have to deal with theory of
Mind in particular second order theory
of mind and talk about belief of that
another may have about the belief of
another etc etc and so on it's it's
become very challenging for the mind so
now regarding um the other question
about the now in the spacious present so
in a way it's a hard question to answer
and we are still working on that in a
way U there is the physical time let's
call it that way which is point like
probably that is involved in the
underlying processing know whatever it
is based on the brain or based on some
computational algorithm so there is a
there is a a timer and then there is the
time as it is integrated perceived and
structured in Consciousness and the idea
here is that because we are doing active

inference constantly uh anticipating the future to some extent even a very close future it can be at different scales so then we are trying to recover from that but we are not yet there the notion of this kind of extended present so that basically at a given time instant uh that's not the right at a given moment if you wish you not only have this instantaneous representation of space in perspective but you have a process in which uh the anticipation uh Drive the next instant in a manner that is integrated ated as a whole so sorry it's very abstract but I'm saying we're also trying to understand how we can put time pressure within this free energy context and we have solutions for that and also how the agents can dive into their autobiographical memory under the same process um we are also very interested it's now it's moving away from that but it connect to questions that we had before to the I would say pathological transformations in the sense that here we are talking about very adaptive behaviors very tuned to sensory motor contingencies and constraint and uh so that has this embodiment the fact that you know we tend to have this first person point of view refer to the representation that we have of our body in space that's constraint that embodiment constraint somewhat reduce the number of possible projective transformation ations and you know because only some of them will give

you that type of point of views but if you relax those constraints then you can get very wild things that I find are very interesting to connect with the literature on psychotrop on mystical experience and arts and to give you an example we not published on that yet but one of the things that happens to the agents is that you know if they they have the possibility of basically being Freer in the application of the projective trans information they can use for instance in the context in which they are in a desperate world in the sense that everything they can think about or do as an embodied agent will yield a very high level of free energy there is no good solution then one option is to release the constraint on the projection and when you do that in order to minimize free energy the agents will tend to basically put their point of view quote and quote at Infinity what we called the god point of view which is a bit bad choice of words and so basically that means that one way of minimizing free energy for those agents is through releasing those con normal constraints is to escape towards Infinity so that basically the world become very very far and then the structure of the representation of the world you have start having very odd symmetries that may remind us of mandas or things like that so I'm shutting up now wow lot of stuff there let's go Adam and then Lee and then

Stephen um hi can you hear me yep sounds
good um
so one thing I've been struggling with
in terms of um time and Consciousness is
uh whether it's better to think of it as
like a series of kind of discreet
snapshots or more of a continuous flow
I'm personally leaning more towards um
discret snapshots and so with respect to
um uh projective geometric modeling I'm
wondering whether uh the refresh rate um
might be like something like Alpha
frequencies like 10 times a second to uh
primarily like posterior cortices
getting into the mix of establishing
that uh 40
perspective um but then more like you
mentioned like access Consciousness
earlier where that might start and like
the whole active inferential unfolding
that that might be more at like Theta
frequencies like three to four times a
second and then like you might get
something like uh northoff style like a
spatio temporal alignment based on like
your interceptive States maybe more at
like Delta frequencies and so you might
have this like nested hierarchy of time
scales with a core screening over events
at various uh resolutions but at at the
bottom of it though um it seems like
where like the core of it is this
establishing of a field um with
projective geometric modeling at the
center and I guess is this are these
estimates I'm wondering discreet do you
think or is it
continuous that's really a very good

question and very sound and interesting hypothesis you know with the empirical testing course it's it's a whole program of research I've been for long an electrophysiologist i work for long on this stuff I I don't anymore but you know I'm very open to pursuing collaborations on that

um I first of all I should say I don't have an answer to satisfying answer to that question um I think it's indeed multiscale um I think that uh it has a discrete quote unquote sharp if not discrete component uh which may be perfectly embedded in a more at least in the way it manifest itself Continuous Flow but that is I don't think in of itself doesn't pose a challenge a theoretical challenge it's just complex but we can think about how it could be done and we can also think about how it could be calibrated based on the type of biological rym you are talking about what is more of a theoretical challenge is this notion of spacious presence in which somewhere that's the words that the phenomenologist like us would use we pretend pretention this notion in the now there is

somewhat Us in the center of that now and there is at the same time a protention towards the future and somehow there is

simultaneously as a manifold an experience of the you know here and now and an experience of the just after and at the same time they still this Resonance of this retention to use us

world word sorry uh retention of of a
immediate past so you can think okay
there is an immediate past there is a
actual present and there is a close
future and then those things are
somewhat bounded together but the
difficulty is to try to think how
actually you can find a mathematical
expression of representation of that in
which they're actually part of the same
whole and not just you know something
you could argue well but then that means
that you know there are ultimately
separate process so we still trying to
work on that see if there is any way of
thinking about that but we are not yet
there cool um we'll do Steph Lee
Sasha yeah just building on on the idea
that the projective Consciousness can be
released when things are not certain
enough to be able to know how to make
sense of in the world it sort of ties in
with this um way that indigenous
approaches are
shape-shifting and they're trying to
attune to an environment which they
aren't able to construct in the way that
we tend to construct and place ourselves
in the center of so I think there's an
interesting thing as well as there being
the
risk just be the the increasing level of
ambiguity and uncertainty in the world
world so that's interesting that you
take that I'm just wondering if there's
as well as the projective thing going
from the self and maybe to Infinity do
you think there's like there might be

another process another type of geometry
coming in the other direction like it
could be and and and and this it sort of
shifts towards the self centered and
there's an
allocentric mechanism as well so that's
you a question and attached to another
work that we are pursuing with
mathematician Danielle Bena and KET Wily
for who just told me unfortunately he
could not make it today sorry uh about
what sometime people have called
prereflective
self-consciousness it's a bit along the
line of this bizarre sense of the
spacious present but now it's about
self-consciousness in the sense that
that's what people have called sometime
the self-consciousness hypothesis not
everybody agrees upon that that we are
always Consciousness is always conscious
of itself that is it's not like a higher
order Theory where there is a
Consciousness for instance perceptual
consciousness of something and then you
reflect upon it and you know become
aware of the fact that you are
perceiving that the idea is that it's
actually the manifold itself is such
that you know there is this
pre-reflective SL
reflexive uh
relationship that is fundamental into it
and so philosopher like SRE I
believe and others I've talk about the
how to say that in English the
um the seers seeing or seen the the seen
seers if you wish the the one we seeing

is also seen at the same time as if
there was some sort of duality in which
whenever I have a first person
perspective and I'm looking at the sky
it feels like or there is something in
it that feels like that the sky is
looking back at me well as it turns out
it's a natural property of the most you
know bizarre strange and interesting
aspect of projective geometry which is
not just you know it's a very complex
and Rich
geometry uh and we're trying to work
that out thanks to this mathematician
around this concept of Duality
reciprocity uh in which there is at once
potentially we still working out the ma
uh uh this the fact that you take a
perspective in implies that you know
there are reciprocal perspective and
that you know you look at something and
these things these things look back at
you uh in your I know it's odd it's
bizarre but you know it's my my attempt
to try to touch upon what you are
talking
about okay Lee then Sasha Adam thanks
for all these cool
questions uh yeah so um I'm anticipating
that I might be about to ask a stupid
question but I'm I'm going to go ahead
anyway so it was um it was stimulated by
the fact that you were saying that you
were kind of interested in you know like
mystical uh experiences in relation to
projective geometry and and I was
thinking about the um uh the meditative
experience of of of

nonduality um you know which is um I was wondering whether you have come across this and whether you see it as in any way related to um uh projective geometry perhaps yeah so I used to be I mean I was at some point I don't know if you know that person a student of Francisco Varela unfortunately passed away during my PhD that's why I moved to the US to work with Antonio Damasio to finish my PhD and

uh of course with Vela who was very interested in let's say Meditation in a broad sense of the term um both for a scientific standpoint from a scientific standpoint as a possible method to study Consciousness but also more at a personal and existential level you know

I got more acquainted to that I practiced it even studied it a bit uh and I've been for instance involved in a very nice Workshop in Israel and perspective on Consciousness where this type of tradition multiple tradition we are meeting together and we had very interesting discussions

and thought and about about that very nice group so um can you try just for me and for the audience perhaps to specify the phenomenology the the aspect of the experience you are

describing the meditator the specific meditation experience that you are describing this kind of unified experience sure I I can try um so I

think in in some sense there was um

uh prior to a certain point there was an

assumption that there's a separation between the Observer and the and the observed and and that starts to break down such that you kind of it becomes it becomes one one space for want of a better yeah so that I think that of course I'm not saying that's the answer I'm saying that perhaps that contributes to that type of state that connects to the what I was talking about of that intrinsic to this the reciprocal relations that are embedded into this geometry I wish the mathematician I work with was here uh there is something like uh it's nontrivial that's important a reciprocity between looking at or aiming at and being looked at or aim at simultaneously it's like it's not like necessarily you lose track of where you know of of the perspectives that you have but there is this more the Richer experience that is implied by this type of geometry so in that sense in that sense at some level you know you you can perfectly conceive within this geometry how you know perspective can be taken from anywhere and are all connected through projective transformation but even deeper uh perhaps through Notions of dualities that I unfortunately could not expand upon because I would you know totally fail that's why I need to work with mathematicians uh that gives you this sort of you know unified experience of a you know sort of some sort of bizarre panopticon where you you know you could

be anywhere you could be looking from
anywhere and from your particular
perspective and being looked at that
could be you that could be something
else and that's you know a manifold of
point of view that at the same time is
grounded into a unique point of view
that exists in full Dimension so I know
it's sounds Des but that's just the math
that's where they where they
are thank you oh we have a new visor but
Sasha go

ahead then

Adam um yeah uh I'm just thinking about
this whole time perception question um
whether and I guess it's it's two
separate things it's is time discreet or
continuous and the second question is
time perception discrete or continuous
and if it is possible for uh people to
have different you know spatial uh
understandings or perceptions then um
it's also possible for people to have uh
different perceptions of time passing
and uh different rate of how everything
is happening and is that um how do we
integrate that or allow for that to
happen that people perceive time uh
perhaps very
differently well that I don't think in
and of itself um is
um

deeply difficult conceptual challenge at
some level at least uh in the sense that
it's not different if you want to think
about it as having systems that
have internal model of beliefs and
understand in the that you know somebody

else may have different beliefs in my robot in the example I was giving it was many beliefs about whether or not an agent likes or dislikes something but uh uh uh you know you and all this let's put it that way let's be broader in the mechanics of the whole system there are a lot of parameters uh that concerns beliefs uh that concerns the the calibration of projective frames that can be different from people to people so I would guess that it's not a satisfactory answer I'm sorry but that could be the same for the the way the the flow of time is U uh structured is is construed uh within Consciousness is to me you know at some point you are living with complex systems that have you know statistical models with a lot of parameters so it's perfectly logical that if you have parameters pertaining to time Consciousness uh they could be different from one person to the other you know at that level of abstraction I don't think the the problem is a is a an essential challenge in the sense that oh here there is a big Paradox that I cannot resolve in a consistent manner with mathematical thinking but of course there are much deeper questions and this whole what we were talking about for this whole spacious present this kind of extended experience of experience of the present of the now is more difficult to grasp to me yet let's go last comments here from Adam and then we'll welcome

Ken hello can you hear me
yep um so and talking about um this sort
of reciprocity or mutuality of like when
you look and then looking back at you it
it started to make me wonder about um
the extent to which uh projective uh
geometric modeling might be uh inter
subjectively bootstrapped and that like
it both enables um theory of Mind
capacities in terms of giving you this
flexible transformation between
perspectives which would be like at the
heart of of like mentalizing a theory of
mind but also to engage that I'm
wondering if that helps to induce the
capacity which then feeds back and makes
you better perspective which then and
and the kind of bootstrapping process
and now I'm wondering in terms of like
Evolution and development like when is
this starting to arise like you could in
theory think of this happening like
nonin subjectively in a way where it's
like I don't know like a predator sort
of like hunting maybe if I go here and
doing sort of like where do I need to be
to get like if the organism has like a
model of itself of some degree of
temporal depth and it's trying to do
something a little bit more labyr and
trying to like Forge for information
well that's go okay I go here I'll have
a different perspective and this is like
influencing the policy selection but um
in humans it's probably really heavily
inter subjectively bootstrapped and know
so some
thoughts I'm not going to make a long

comments to not take too much space but
I find everything you said I agree with
you I think actually you right probably
so uh there are very interesting stuff
in ethology and you know animal behavior
and even in recent relatively recent
literature that you know suggest the
role of projective representations into
the way
animal try
to lure others into you know either
being a ma or you know being a a
pre cool
welcome Ken you
can say hi yeah sorry I'm so late all
goodful morning do you have any uh thing
to add or just feel free to raise your
hand any time okay we'll do cool thank
well there's a few more slides this was
really a great set of questions by the
participant so that's really fun topics
uh here's some quotes that I put up
there and the part I wanted to kind of
start off with is the second half of of
the uh second quote on here which is
just uh a system running the PCM would
more closely approximate the actual
complexity organization and processing
power of human or animal brains and this
could very well on the author's view
literally be an artificial Consciousness
so maybe getting into sort of questions
about what does it actually entail to
have the awareness what if you make a
system that has action and perception
and it does a projective geometry
transformation and it you know it's
doing it because you're the one who made

the robot so I was just curious to hear about from the robotics side or from the mathematical side or from the neurophenomenological what else is happening in humans is it evolutionary is it linguistic

Etc so I I would like Ken to to speak because he he has a lot to say uh but since you ask things about Robert in that question is very relevant of course I will answer as a scientist I'm hiding behind the method the scientific method not to avoid the question but to tell you first that I don't have an answer to that fundamental question what I'm trying to do and that's I think the only thing science can do at least as we know it and you can take the model of physics as a prominent exemplar of that is to try to understand not necessarily why but how we do think we do things and that applies to Consciousness to the phenomenology of Consciousness so what we are modeling here is that phenomenology and how it is involved in other process and so at the end the only thing I could say about my robot in the way they are implemented at this point you know there is a lot of work ahead is that at least their behavior will be governed by principles that are Central to the way Consciousness process information let's put it that way so it doesn't imply at all that those robots are actually conscious in the sense we we would like to um but at least uh uh they can demonstrate how

our Consciousness and the principles that seems to shape our Consciousness play a role in overall the economy of our Behavior etc etc it doesn't give us a an answer to why you know this is conscious in the sense maybe actually it's with a certain level of complexity sufficient and maybe Ken should speak now uh and rebound on that well I was going to say this um passage if uh memory serves we we we of course this is a a general sort of worry that anybody who uh works on Consciousness thinks about it has especially if you are doing anything with u simulations as David was so we started to think well of course we don't think that um the simulations that David has built are actually contained Consciousness you actually contain conscious agents so then you wonder all right you know so it's not it's not merely um processing information in the the way in a way that simulates what's specified in the model I guess you could say um it could be other things as well like uh so in a sense we we think okay this is a necessary condition for Consciousness that you process information and the way the PCM outlines um sufficiency require a certain kind of uh scale and complexity that that we we're not we're not certain about yet so we shouldn't just um assume that any any kind of that that uh follows this this model is just ifof facto a conscious being that's kind of claiming too much

what what else is involved well we we
don't we don't know but um it it seems
like uh this is we we we maintain this
is necessary for Consciousness so um
conscious being will will have this
structure well will process information
in this way
um and for all we know it it could be
sort of
um things that we don't we're not going
to be able to reduce
to uh principles so easily that we'd
have to discover empirically that's the
that's the reason we say here the an
aposteriori identity Theory that's
that's the idea that you can't really
say what Consciousness is AR priori you
have to find out what it is uh
empirically
and at some point you might find
yourself in a situation where you could
say Okay
um these suppose we build a lot of
artificial systems that do this these
exhibit all the behaviors and so on that
make it seem like you know a conscious
being um they you know suppose they
operate on a certain scale and a certain
complexity at some point we might be
able to say yeah we think right here's
The Sweet Spot why exactly we don't know
maybe we will discover that maybe not
but um on the one hand uh it seems like
you don't want to
be um shy of saying that the you know
from what we can
tell these are necessary conditions and
on the other you don't want to be to um

you don't want to you don't want to have
a view of the sufficient conditions for
Consciousness that make it
quite too easy you know so awesome yeah
thanks for that response Stephen just a
really quick comment and then we'll move
on just just quickly I was just
wondering when you think about
projective Consciousness is it like this
is really summing up a particular type
of Consciousness which that we we are
taking these projective
perspectival kind of um ways of engaging
the world and there's there's a
potential I wonder if there's a
potential for that to seem like it's
more like how Consciousness is because
we culturally have taken that on to be
how we think of the world but this might
be like one way of taking a view from a
more sort of higher
perspective so should shall I take this
one can please do and speak
more what's that I think you should
speak because I have spoken already a
lot and
you so I mean our our hypothesis is that
this is not just of
um you know uh how we've been
enculturated or uh let see if I I I got
your question right I mean r view is
that this is this is a at least a
necessary feature of all Consciousness
that's that's the
claim so now you could imagine and you
could you could actually wed this kind
of view to other theories of
Consciousness you could wed it to you

know pans sism for that matter you could just say oh no actually Consciousness is something else it's whatever some fundamental property of nature um but this then if you had that sort of view this kind of work would still be of interest because it might um sketch out how cons ious beings like us or probably you know animals um how they experience the world and how they organ how they process information all right so but and of course when we make it so it's compatible with that comptible with um really of Consciousness that's cast at a kind of metaphysical level I guess um our rvw would be something like well you know suppose this suppose this type of model ows us to explain everything that we we think is pertinent to Consciousness so it allows us to account for whatever behavioral and cognitive functions um Consciousness carries out and for the phenomenology of Consciousness um if it if it allowed you to to count account for those things um then and you suppose you could identify the type of structure this is and the the sort of scale and complexity it needs to be organized on then to our way of thinking at least as it comes to the metaphysical issue we're we have a tendency to think okay that's just what Consciousness is we we discovered this empirically we discovered it aposteriori there's no need to postulate any further metaphysics because any any metaphysical account of Consciousness will actually

has the same structure it's it's
something you didn't figure out a priori
really you just thought okay well here's
a here's an account that would make
sense of the fact that you know say take
pan psychism why is anything
Consciousness well you conscious well
you know if everything is conscious then
it's no mystery that we
are right so so that's uh so that's
that's an argument you could make well
you know as you as you probably know
there are all these different um
positions in the philosophy of mind
about the nature of Consciousness and
the and the distribution of
Consciousness that in my view they're
all kind of epistemically On a par none
of them are really like you can't show
any of them a priori you can't
demonstrate so um it seems like it's
wide open to you on the theoretical
level to be this kind of identity
theorist just when you reach you know
when have this structure have this
level that you're a conscious being
that's what Consciousness is we figure
it out empirically and we don't need to
sort of postulate anything else that
said it's totally compatible with with
other views so you could hold that not
this is much more derived and primitive
like sensing or something is is is a
basic form of con or something like that
um we we do think though that e so
here's one thing that's it's not exactly
AR priori but let's say it's a a strong
intuition that a lot of people have if

if you're conscious of anything that's
going to involve the appearance of
something
to so it has this kind of appearance of
to structure
and that's a kind of spatial spatial I
mean quasi spatial structure a protog
geometrical structure um and and that
takes you already kind of in the
direction of of thinking of
Consciousness as like a space and like a
projective space in fact right um that's
not you know this you can't demonstrate
that a priori but it is a strong
intuition that people have so I'd like
to say I think I say in this paper don't
I um you know follow your
phenomenological
Muse build your build your models and
see where they take you because we can't
really determine those things a priori
or just by banging on the table with our
our favorite
in thanks um I I see Adam and Lee have
their hands raised but just really quick
would uh especially uh David and uh Ken
would would either of you be okay with
staying for like a few minutes past the
hour so we can just continue talking a
few minutes okay
I have I have a call in about 40 minutes
but so I'm fine oh beautiful so we'll
just um anyone can depart whenever they
need to and uh otherwi we we won't have
to come to a speedy Landing there but
we'll go Adam and then
Lee oh hello
um I guess I'd like to hear a little bit

more about uh the intuition for or um
why it is that existing simulations
might not be conscious and in terms of
um what it would actually take to uh get
a sufficiency um this is something I've
been struggling with also um the uh for
me like I've been like mostly coming at
it from like the neural Direction
thinking of like okay what in terms of
neural systems might underly these
different capacities and then like what
uh how would that be implemented um I
become a little bit more confused when I
start to think of alternative
implementation so like something like um
uh projective geometric modeling I'm
wondering if that's primarily like a
posterior meal C posterior medial
cortices and that's like giving you this
sort of flexible spatial modeling but
then this is like coupled to like um
body maps of like lateral parietal loes
and that this is part of like one um
basically synchronous complex which is
mutually constraining each other as
you're filling in this sort of joint
estimate of the world and your lived
body in relation to it so basically
exactly what you propose um and so for
me it's like when I'm talking about a
biological system there's a certain
grain at which this calculation is being
established through like a metastable
synchronous complex or might call it
like a non-equilibrium steud state
distribution or I call it a
self-organizing harmonic mode but if
we're talking about something like a

simulation on a turing machine um that
might be doing the geometric modeling by
very different means um is that an issue
is that not an issue for you uh just
curious to know more thoughts on uh
boundary problems thanks maybe I can
give my thought about that and then
Canen number one about the functional
neuroanatomy and the
neurodynamical mechanism that you know
may implement the type of process we are
talking about I think agree with you in
the paper we published in 2017 in the
Journal of theoretical biology we have
at the end a a little figure that kind
of summarize or gross functional neuron
anatomical hypothesis and indeed it
seems that the posterior you
know in particular the and to some
extent the poal cortex but the postal
cortex may be more is a bridge H towards
interaction with other system but the
paral L and then we could enter into the
detail uh uh is certain is probably
where this sort of project as we call it
in the paper projective geometrical
engine operate and I think the way you
you attempt to make sense of the
difference between you know something
like the model being processed the way
you describe it into a brand
versus into a touring machine basically
a computer is is is perfectly fair and
right and maybe indeed when you do it in
the touring like way like when I do my
simulations with my virtual humans and
robots you know it doesn't you know unil
the type of experiential aspect that we

are trying to figure out of which we
are trying to figure out the cause
basically the sufficient condition and
perhaps in is it's a very different
context of processing the way you
describe it and that is actually key
perhaps not you know it would need more
elaboration of course there's still
explanatory gaps here but uh to make
that process conscious so for me it's
not a problem it's different type of uh
of implementation we can use the touring
life to do simulation from which we can
learn a lot but probably inde there's
something about you know the embodied
processing with this type of what about
the brain is doing and you know I'm a
neuroscientist initially and I can I we
all know that actually we we know much
less about the brain than you know we
pretend to uh and there are probably
great and important deep fundamental
principles still to understand and until
we have understood the brain in a much
deeper way uh um well we we won't be
able to con on that
question Ken yeah I I can just D that
you know um I mean suppos it it's it
sounds a little odd you know from the
kind of computer science perspective to
think that say doing performing all
these computations in a massively
parallel way so that um you know I mean
you can imagine you know a tinker uh
computer doing the computations involved
and taking like thousands of years or
something something and you think okay
that's it's not it's not merely running

this program right but if you say okay
it it has to do with um massively
parallel integrated computation of these
uh functions um that of course sounds a
bit like um like strange like why why
would simply being able to do it really
fast or do it in this parallel way make
it make a difference um but you know if
if you're serious exterior identity then
you think well okay we'll just find out
right it it does it does seem really
strange though to think um that a
touring machine made out of beer cans
and string as John

Sur carrying out

these would be Consciousness so um so
that's why I I think it's it's a good
idea to leave leave a little bit of SP
here

right so that's which is what we to
do maybe let me just add a little
historical note it's not about touring
but it's about vanon nyman you know in
his seminal paper in when he was at the
Bell Laboratory I believe and you deriv
he explained the foundation of his
architecture that is now basically a an
implementation of a touring machine into
the computers that we we use currently
um actually and then you know it becom
like oh you know vanman Machine versus
the wet brain but in fact Van Norman in
this paper start from models of
artificial neurons the basic mular canit
neurons to try to figure out how basic
computation such as you know arithmetic
operation could be abstracted and then
he proposed a Serial model of computing

emphasizing that at the time because they had to use those vacuum tubes to make those the equivalent of semiconductor you know it would be too too big to build a parallel machine it would just not be physically practical you would need like a giant building just you know to put enough tubes in there and so the solution of doing it in a binary Manner and in a kind of sequential manner it was just a practical solution okay but his thinking was broader so you know maybe there are also deeper aspect of computations uh you know we don't understand yet and that could help us also to make sense of the experiential

Dimension that's an hypothesis and if I could add one thing that it is also true that simply massive complexity hides I mean it it has a sort of I don't don't take this too seriously but you all know the old sort of Hegel's idea that um quantity eventually yields to Quality so you have a quantitative changes and you have enough of them it's as if it transitions to Quality well there there is one way of thinking about that um in a kind of not not taking the metaphysics of dialectics seriously but looking at it uh epistemically where very very complicated things can end up being in fact this is kind of the way we are uh oh somebody brought this up okay okay well all right so I'm speaking to nobody's gonna be freaked out if I start bringing up

go know um anyway that the idea is yeah
that particular notion of of uh going
from quantity to to Quality so just
think of massive
complexity you take a mass complex
system as a as a as a whole it can
behave in ways that you you have to
understand sort of
qualitatively um you can't I mean you
know the the
even things like CH chess play computer
programs to take the classic example
approach your chest bang playing
computer if it's a good program from the
intentional
stance Dennis term you you you regard it
as you know trying to attack your and so
on you don't like think of the low-level
complications going on in it to to
implement that so that that's all just
to say massive complexity with the
parallel massively
parallel it it's not totally crazy to
think that that um could yield something
that has this kind of form and this kind
of macro structure if you like that we
experience
qualitatively that Consciousness has
it's not completely crazy to say okay
yes serial computation couldn't do it
but massively parallel
computation could here it's no crazier
than a the of Consciousness yeah go I've
been
um thanks for that I've been um
that makes a lot of sense to me
and I guess in my own thinking I've been
struggling with like to what extent I'm

dealing with practical versus in
principal uh uh issues with respect to
like boundaries and like what it would
take to realize Consciousness so like
you say like the massive parallelism is
practically necessary but is it in
principle necessary and like and there
does seem to be something strange like
you know Church turning uh thesis like
in theory you could do it with the beer
cans and rope and let it just play out
in a really long time scale and
eventually you calculate the whole darn
thing right but you're just like I don't
know about that but but why so it would
be like well practically there might be
like a sense where like okay you need
Consciousness to both like inform and be
informed by action perception Cycles or
something like that but that's still a
practical not necessarily like an in
principle but then like I I guess I've
been wondering like okay so if
it's we're going to in principle would
be

like is the computational objects like
real or not um so like uh if like like
like a simulation versus an emulation or
things like this so like in like like in
with like the recurrent Dynamics sort of
like metastable synchronous complexes
model I'm working on uh
even then there's like an issue of like
is this a real thing or not and like
they're dealing with like this issue
with respect to non equilibrium uh steud
states in the fore energy principle
which is like these um these things are

defined as these like harmonic modes um
but
then their boundaries are like
um they seem to be um model dependent
like like what um or like an neural
system like do you ever get like exact
synchrony how much is enough to count um
what's the
uh what's the time scale at at at which
the thing actually has its existence as
a particular attracting State and
relative to what and so this is where I
start to
uh yeah this difference
between practical and in and in
principle issues of impementation and
what would
actually and then what would let you say
okay this thing is conscious I'm I'm
still really confused about it anyway
may I interject well ju do me just say
may I comment on that yeah I'm gonna go
if you com then you I don't seem to have
a opposition so I started developing an
hypothesis a while ago um that I
published in you know in a journal
called advances in Neuroscience that
actually died it I don't think it still
exist uh and it was kind of based on a
reflection on my own work and the work
of other in neuroscience and neuros
psychology notably working with uh brain
damage patient showing that you know a
lot of what we believe as being critical
for cognition in many aspect of you know
cognition and effective processing in
the brain like everybody believes now
that the insular cortex is critical for

craving and know many aspect of the sense of self well I worked with brain damage patient that basically had no insul on both side lack 40% of their brain and nevertheless you know except for some deep anog Amnesia that could be explained by their whatever uh uh they were normal so I came to actually think that the brain is much more complex than we think that there are the way we observe it with the method we have no Imaging electrophysiology is very low level it's not different than putting omers on the microprocessor and trying to infer what type of operating system running and so I started thinking so maybe there are you know layers of abstractions that we haven't understood in the way the the brain process information and one analogy that came to my mind I was very interested in computer science at the time is the process of virtualization in computer science that is basically a way I mean it's a simplistic way of explaining it but uh a way of creating abstraction layers so that actually at the user level of course we're talking about computer involving user but at the process level uh uh all the details of the hardware are hidden you know when to take a very trivial example that is not a good analogy when you use your desktop computer you don't see the the the machine language you are not using word you don't need to program in Assembly Language to use your Microsoft word uh interface so virtualization is actually

a process through which
um software or process can be uh
abstracted hide the complexity of the
hardware uh and also become much more
resilient in the sense that then you can
kind of move things around across
servers across you know different things
so it's not an answer to the question
but I I still have this intuition and
believe that perhaps there is something
analogous that is Bridging the Gap to
some extent between you know what we
know and observe of the brain and
subjective experience and more you know
higher order level process that is
analogous to virtualization that
basically the phenomenal experience of
Consciousness is very useful for active
influence Etc but relies on neural
mechanism that precisely are designed to
uh hide the Nitty Gritty small and
complex details of the hardware and
there and therefore makes it more you
know usable more practical and of course
at the same time it takes longer because
you need to run many cycles to do that
for instance if you want to S simulate a
certain type of microprocessor over
another architecture of microprocessor
etc etc so I think there's something
there about trying to understand this
how the brain May create virtual space
and through virtualization process that
then may help bridge the gap between the
different domain we are talking about
even though it's not sufficient yet in
my mind it doesn't fully address the
problem just a couple notes to um tie

together a little bit of the discussion
so we've heard about a few classic
philosophy of science stories like
necessary and sufficient uh practical
and principled from philosophy and then
also like real versus unreal seems like
to some degree brains would like have to
because they're so complex they would
have to create these abstraction layers
just to like work as Control Systems
just to work as Integrated
Systems they'd have
to yeah like you'd have to like self
core science just to actually have purchase
to do like heterarchical control and
then once you're doing this um then you
might get potential like flexible
multiple realizability from that
potentially or or or or non flexibility
too depending on like how you did it
but yeah you summarize Adam again very
good summary of in better presentation
of my home
thinking well you've uh influenced me
very heavily so uh I don't think that's
a
coincidence um yeah one just to bring a
few of these threads together because
it's all really interesting stuff so one
is that in complexity we say more is
different and that's related to this
qualitative to quantitative uh more
quantitative changes can lead to
qualitative differences through phase
transitions self-organization just
qualitatively different Behavior strong
or weak emergence from just
quantitatively more and then that's

always going to be a relational question
how many grains of sand are in a heap
that's the whole point is that's not
pretending to be an objective question
it's about object but not pretending to
be a non-relational question even though
it is about the world uh and these ideas
about which levels of the system are
available for introspection
versus is related not just to attention
schema Theory and attentional balance
salience from the Fe and active
inference but also a lot of things in
Gerd Gigerenzer and so there is a
Hofstadter citation in a lot of these
ideas have a lot of analogy there which
just shows how in some senses many of
the same questions are returned to even
40 years later so we have to balance out
the big Computing hype from the really
quintessential philosophical questions
like necessary sufficient practical
versus principal real versus unreal
perspective and taking that seriously
the relational perspective integrating
topology and geometry so it's just
really awesome to hear this great
discussion and that the way the paper
and actually the sequence of papers is
written is very clear interfaces and
highly compatible focus with different
metaphysics but just saying it's neither
uh simply necessary nor sufficient to
have this projected mapping it's just
under condition X and Y for example The
Vanishing um you know privilege point
and the all these things we were talking
about it just goes Downstream from there

so it's an interesting approach and as we're hearing from this discussion a lot of that come together in thinking about these topics so thanks for like bringing all these perspectives the tables the school stuff um maybe Ken can comment on your reference that you had before on the glass of St works and get and bar because actually Ken has worked a lot on these questions has actually contributed to the I'm a strange look strange Loop book of the glass of Stater by editing a version of it and he knows very well maybe he can things to say cool this is one the this is the section that where it was one of the sections where hoffstead was cited but I'd love to hear about this Ken it' be great thanks yeah well it's just a little bit of U I guess history about this like um actually I met David through um he he and Antonio deasio were contributing a paper to a book I was editing um and Douglas hofstead also had a paper in that book and um you know the papers were supposed to be be like I don't know 30 pages maybe and hoped or turned in 100 pages so and it's kind of typical way so I I uh I said well look you know we have to edit this and he's like you know um okay but you know be very careful and so I went through this long process of editing down his contribution I cut it in half and and that eventually became um I am a strange Loop which is cited here um but he he was uh ultimately okay with with my editing job but that's yeah

so and and in particular I I what I
thought was really ended up being really
nice about um I am a strange Loop uh the
way in which Hofer is able um
describe you know the ca basically the
causal efficacy of um
imple you know of implementing
mathematical forms and so it's it's it's
you you you know you T at least in
philosophy think think of mathematics as
a a totally like abstract platonic way
it doesn't have any causal powers and so
on and then of course there's the old
much you know there is the Aristotelian
tradition which says no no the forms are
actually in the matter they're
implemented in the
matter and but that's that position is
not particularly um I mean you can find
it but it's it's not kind of like it's
not the dominant sort of thumbnail
sketch of this issue in the minds of
most philosophers but I found Hof's book
to be really um really good on that like
you know implementing certain
computational structure certain
ultimately mathematical forms has real
causal power you know it does change uh
the system that implements it and and
and it's perfectly legitimate think of
it as a causally real uh thing yep one
example from hofer's writing that
reminded me of today was when we were
talking about whether it would simply be
enough to have a data processing model
that involved a projective geometrical
component would that be necessary or
sufficient for Consciousness and your

very articulated answer was about how actually we know that these are some sort of uh features of what we want to be modeling but they're neither at the necessary nor sufficient tier they're things that we're seeing as like sort of being spun off from some underlying process that there's still a lot of mystery around and so that's like the phonographs um with the low resolution photographs in go to leure Bach it's like you can make a very limited thing that does one thing without error or without some of the fancy attributes but then to extrapolate from this less is different more is different case it's like well it's not a phonograph it's a hammer it's like okay it doesn't play records yep well it's that kind of a phonograph that's what it's like to be that kind of a phonograph that's what it's like to be a light switch okay okay well that's not really the complexity of system that we're talking about here which is the embedded in culture human and then that level of complexity the full featured phonograph opens itself because of its strange Loop nature to this whole new universe of recursion and of loops and of uh self-reflection projection and then the phase transition or more is different question is like how do we get from things that don't have that strange Loop feature to things that do but there's nothing that is sort of a strange Loop or is not a strange loop it's like either it Loops back with sufficient complexity to be strange in

the world or it's just not strange
enough to be as interesting as these
open-ended infinite generative systems
that we seem to be participating in
Adam and I'm curious about how this
interfaces with free energy because I I
don't think there's been a hset or free
energy crossover quite like this
Adam hi um I'm thinking again about what
uh David was mentioning with this sort
of like reciprocity or mutuality of
perspective of like seeing and then like
looking at something and then this and
being looked at and the ways in which um
some of these things might be inter
subjectively
bootstrapped and I guess I'm wondering
um whether that puts like a strange Loop
like right at the heart um of this model
and I'm also wondering whether one of
the things that might like potentially
get you is um recursion and like and a a
powerful kind of
reflexivity which would then you can
then start talking about things like
conscious access and all sorts of
metacognition kind of like growing out
of that potentially um yeah some
thoughts K you need to answer that
question you know in reference to self
reference and nonwounded set theory and
whole project Etc you are the the one to
to address that okay well so actually
that volume that David contributed to
and
deasio well deasio but Heder contributed
to it was it was a book called
self-representational approaches to

Consciousness um and it's was most of it
was written by you know analytic
philosophers and some some couple of
phenomenology phenomenologists but the
that was the theme basically how do you
how you know people who think that like
like I long have that Consciousness
involves this kind of
reflexivity kind of self-awareness that
is not just it's not just introspection
it's a more basic notion of reflexivity
than being able to attend to yourself or
reflect on yourself and that that book
was was about that but uh in that book I
kind of looked at ways of formalizing
this using so the Hop the hopare we we I
asked Heder to contribute to it because
we we thought that's that's right in
line with ger leer Bach and the notion
of kind of using the notion of
metamathematics and self-representation
in self reference in um in the
metamathematical context as sort of one
model for the kind of selfawareness
that's involved in Consciousness he of
course that's the Str that's the kind of
quintessential strange Loop for him with
respect to Consciousness um but we I you
know looked at models using hypers sets
like sets that contain themselves or as
members um
and now recently though I mean we we
kind of actually working on that is kind
of at the origin of the the whole
project the whole PCM um project because
um I I moved back to Iowa City at in
like
2007 I think that's right and um David

was there um at
uh still in the in the he had um
graduated from the Neuroscience program
there he was still working at the
Neuroscience program at Iowa and we kind
of formed a little research group just
based on this and this eventually you
know sort of we we could of discussed
geometrical models a little bit but but
David really had the
Breakthrough um with regard to thinking
of projective geometry as the the the
Way Forward U somewhere around 2010 or
something like that but we um we long
wondered how how can you you know can
you really connect these two threads
this the the geometrical part um the
projective geometry and um this notion
of for lack of well to use Hof's term
strange strange loopiness of
Consciousness um and we we're actually
still working on this so we we we we uh
we think we're not certain but we think
we thanks to Daniel because D Daniel
benan um we think we we found a way to
connect these two Notions finally and
more clearly um D do you want to say a
little bit about
that about so how the how the how the
PCM can sort of um make sense of the
this notion of strange Loop loopiness
strange loopiness yes so that was really
a driving force in in trying to select a
a model to that the model would
incorporate
intrinsically properties like that so
when I was trying to because I always
had been convinced that you know there

was something geometrical about intuition and so we needed to move from kind of abstract graph Theory and even with this kind of non well funded sort of set theory that allows you these Loops Etc into something more phenomenal more experiential more concrete and so I was looking into kind of possible geometries that had weird features uh Beyond this perspectival intuition from which we introduced that and this projective geometri geometry uh under some boundary condition but actually it's intrinsic to it as a natural Loop if you wish what I'm going to say is going to sound very awkward and it's a kind of qualitative description of things that needs to be expressed and formulated ously so sorry for to the mathematician that may listen if I'm not doing it the right way but basically when you look at a point at Infinity in front of you okay so the vanishing point just in front of you that's the same point that's the back of the point that is actually the antipodal point in the back antipodal points so two opposite points at Infinity in that Pro in projective space are topologically glued so that some sense actually if you were kind of a virtual little creature moving into the space of Consciousness uh when you reach the point at Infinity you find yourself on the other side of the space uh and uh if you think about it it's just as if uh actually the projective space you know

at you know this when you kind of start
messing with the or considering the
plane at infinity and it's top ological
properties basically was kind of a sort
of mirror in the sense that basically
when I look in front that's not the case
in your visual experience obviously but
that's in that is in the structure of
the geometry I look quote unquote of
myself looking in front from the back in
a kind of reverse way that's a property
that is actually intrinsic to projective
spaces but that we don't normally
see okay cool stuff here thanks for
sharing all this history too so just two
thoughts one is this virtualization idea
and in gerle user Bach it's Prelude ant
Fugue Aunt Hillary the ant colony and uh
is it a virtual machine well there's yes
and no in one sense lower level details
are shielded to the more semantically
operative top levels so that's
virtualization for sure concurrent
programming as well but is it a virtual
machine in the sense that it's a VM that
I can run on a different operating
system no that's not even the ant colony
OS that's not a strategy that transfers
so the virtual machine metaphor captures
one aspect perfectly which is that
there's shielding of lower levels of the
quote assembly code but also there's
another feature of virtual machines that
they've come to mean in the modern
Computing environment which is like
reproducible Computing and that's
definitely not the case that's because
we have this embodied inactive

ecological element so that's kind of a cool uh extension of a hoffstead idea and then this idea of infinity being on the other side it's sort of like you would if your light rays were being bent around the sphere or uh whatever shape we're on it would be seeing around so that's straight out like a laser so that's sort of an intentional laser reflected by the higher concentration of photo receptors The increased sensitivity to observations there um but that being the point of attention and even meditative practices often focusing on attentional points and uh visual Rays like that but it does if you think about the the ray of light going all the way around it would be the trajectory you would look at to see yourself from the back so very interesting there and that's related to our conception of the space of the earth whereas on a flat plane it would just go to infinity and then pop through in like a piercing way because the space doesn't Bend back on itself

Adam sorry yeah

so I'm thinking that you um might be providing an additional account of some of the uh phenomena that uh gratiano tries to describe with attention schema theory in terms of um certain intuitions about the nature of Consciousness and selfhood and a kind of like um awareness is like a soul like ethereal non-material intuition and so for gratiano for him uh Consciousness is you

know it's this low dimensional sketch
necessarily reduced dimensionality for
the sake of control um to have a model
that's more handleable that throws away
details and gives you purchase but that
this like low dimensionality to it is
like the thing that makes you that if
this is your awareness this is um and
your sense of Consciousness and and self
coming from this that this is part of
certain intuitions about like immaterial
souls and things like this but if your
perspective inherently involves this
kind of like strange Loop this like
difficult conceivability where it's like
you are both the constant context for
everything the reference point but
you're the one thing that's like
invisible and in this kind of
paradoxical way I'm wondering if even
I'm wonder if that potentially
contributes to similar aspects of
phenomenology where like certain asp
of kind of like mystical
intuitions about the self and
Consciousness some
thoughts cool question and we're taking
it nice um and interesting areas David
why don't you go ahead Ken Ken could say
a lot of things oh sorry somebody else
wanted to talk nope you or Ken go ahead
okay I'm G I'm gonna just say something
but Ken will be able to better elaborate
there was a
if a Mystic or say zist named Rudolph
Steiner right K who actually uh believed
that projective
geometry uh because of this sort of

reciprocity of perspective when I look
at the sky the sky look at back at
myself was a connection to
God we don't endorse that view
necessarily but there is an history of
mystical and esoteric Traditions that
invokes projective geometry in the past
in that kind of context right
Ken yeah so there's there's the Steiner
uh tradition which um one of his
students I believe she was one of his
students Olive Witcher she wrote a
really intruction to projective
geometry um which uh I highly recommend
looking at it's got wonderful
illustrations and um and they they seem
to have been on to idea that the
projected geometry was crucial to
Consciousness and mind um and but you
know they also had a lot
of strange other Notions that that made
them seem like cranks you know um and
then there was another um interest in
just higher dimensional geometry that
came from people like um uh PD ensky if
you've heard of this of this this guy um
he um
he was he wrot book called the fourth
dimension and his his view is that there
must be part of our selves that we don't
have access to in the in in the way in
which we have access to
three-dimensional objects which kind of
and he thought that would be in the
fourth dimension it's not clear that he
thought that it was a projective
geometrical structure but um it's still
a sort of you know sort of in the

ballpark um well just one note there is
that uh ensky who studied under uh GFF
there some deep esoteric threads and of
course going much deeper Through Time
alternate ideas about number geometry
color and of course philosophy those are
in many ways what define the uh less
religious at least in the exoteric sense
of these wisdom Traditions so it's very
interesting to see the geometry
spiritual connection return after who
knows maybe never even a separation but
at the very least in most people's mind
they' think of it as shapes on a plane
not about the conceptions of the lower
level structure of reality or the higher
level structures of reality like the
shape of the universe Shape of Things
shape of ourselves so very nice threads
and David's a big fan of uh what's his
name matila geeka isn't that right yeah
poet mathematician and Romanian prince
who wrote many book of about the
connection between mathematic Aesthetics
and natural form yes there are a lot of
connection that were very inspiring to
me my father used to show me his book
when I was a kid and my grandfather was
an art historian and he was also
searching you
know interpretation of art forms through
mathematics so there is a long family
tradition that led me here so I'm sorry
but I will have to leave to another call
it was very very great I can leave you
and you can continue discuss and was
very happy so uh we'll get in touch and
and stay in touch and I'll see you great

bye bye see you I'll um move us to the end but really uh D and Ken thanks so much it's like always just great to have the authors on here and to be hearing all this extra stuff so the other slides just to you know visualize them people could screen capm is just a few more quotes to get to that he hegelian dialectic which we actually got to talk about um we could have had a few more just general questions especially where does active inference fit in into this whole picture and that's in the cybernetics and how it's carried out but uh active inference live stream and whatnot that's always what we want to think about and then we just had figure one figure two figure three and figure four so those are just things that we could have talked about maybe uh in the future or in 2021 we could have the Moon Illusion or a deep dive on one dimension or another because this was such a fascinating conversation so if uh Adam or Alex or Ken maybe each of you want to have just a uh last comment that would be great so let's um Alex if you want to go first then Adam and then Ken or pass no worries if not okay okay I'll pass by now thank you Alex um Adam any last thoughts um how's it gonna change what you're G to work on or think about or just um too many thoughts but one of the things I'm uh I'm doing a lot of thinking about like what neural systems uh might potentially contribute to different aspects of uh uh visual

spatial awareness and geometric modeling

and um

one issue I'm wondering about um is the extent to which

um

uh at least its visual nature it might

always the conscious awareness might

always be um more of of a of a 2d

projection and that um things like

threedimensional so and things like

additional depth and additional

Dimensions might involve um uh the

ability to uh transition between

different points of view in a coherent

way in a way that would

imply a 3D world but where at any given

moment the visual aspect um is uh

actually uh 2D and potentially without

clear depth um these are some things I'm

wondering about and I think it's not

incompatible with projective geometric

modeling it would just be um whether the

um the engine you know how distributed

is and like the granularity over which

it's being achieved and so this like

coupling between like a a 2D projection

screen and like you know a body map like

adjusting what you ought to see or

something like the posterior singulate

like saying like okay here's where

you're like your eyes and your head is

so you know giving putting your

perspective in there and then

putting it seems like 40 geometry could

be perfectly compatible with this and

but I'm trying to figure out how that

works that was a lot of rambling but

thank you no great thanks for your

participation and this figure two I think points out some of what you're talking about that's a projection onto the 2D there's an ambiguous projection on the left because of the perspective being ambiguous and then free energy is invoked as like a symmetry break that gives a perspective on which side of the box one is actually on given a prior that about how certain angles must be reflective of two-dimensional representations so yeah Ken any um thoughts on that and or just a last comments uh well just thank you for organizing this and I'm I'm sorry I couldn't make it for the whole thing um and it's really been a wonderful conversation and I'll just say the uh the next thing we're working on the Daniel and d and I is a paper trying to relate the projective geometrical model uh close or to show how it relates to Notions like strange strange and um pre-reflective self-consciousness this kind of more basic type of self-consciousness than uh introspection or or attention and to show that you can you can make sense of that in terms of the projective geometrical structure that's the next thing on the agenda for us cool sounds like a fun research Direction so with all of this being said everyone thanks a lot for participating live and if you're watching for watching it too we will have the follow-up form link is in the calendar invite if you want to just provide us some thoughts um any feedback suggestions and

questions are always welcome other than
that just uh stay in communication
really appreciate this great
conversation and some awesome stuff so
thanks again everyone for participating
and watching and we will talk soon so
see you later thank you